Genetics of the Microworld

How bacteria switch genes on and off, and how they talk to each other

GOALS FOR TODAY

- Explain what an **operon** is and why bacteria use them.
- Explain how the **lac operon** is switched on only when lactose is present and glucose is absent.
- Explain **quorum sensing** — how bacteria "count" their neighbours — and why it matters.

1. Overview: how do bacteria control their genes?

Bacteria look simple, but even the simplest bacterium has a tricky problem: which genes to use, and when. The bacteria in your mouth and gut have thousands of genes. Most of them code for proteins, each with a special job — getting energy, building the cell, defending against viruses.

Some of these proteins are needed all the time; others only sometimes. So a cell does not use all its genes all the time. Think of the genome as a cookbook with many recipes: the cell cooks only the recipes it needs right now.

Gene control (gene regulation) can happen at many steps, but a lot of it happens at the first step: transcription. Bacteria have regulatory proteins that bind to DNA near a gene and either help or block RNA polymerase, the enzyme that copies the gene into mRNA.

What is an operon?

In bacteria, genes with related jobs are often placed next to each other and are controlled together from a single promoter (the place where RNA polymerase starts). One long mRNA is made that carries several genes. A group of genes controlled by one promoter is an operon. Operons are common in bacteria and rare in eukaryotes such as humans.

Operons let the cell switch on a whole set of genes at the same moment — the ones whose products are needed together.

2. The lac operon

The problem the lac operon solves

Lactose (milk sugar) can feed E. coli — but only if the cell makes three special proteins. Making them costs energy, so the cell only does it when it is worth it. E. coli prefers glucose because it is quicker and gives more energy. If both sugars are present, the bacterium uses up the glucose first and only then switches to lactose. So the lac operon should be ON only when lactose is present AND glucose is absent.

To measure the two sugars the cell uses two regulatory proteins:

- the **lac repressor** — a *lactose sensor* that blocks transcription when there is no lactose;
- the **catabolite activator protein (CAP)** — a *glucose sensor* that boosts transcription when glucose is low.

Structure of the operon

The operon has three structural genes, transcribed together as one mRNA:

Gene	Protein	Job
lacZ	β-galactosidase (enzyme)	Splits lactose into glucose + galactose; also makes allolactose
lacY	Lactose permease (membrane transporter)	Brings lactose into the cell
lacA	Transacetylase (enzyme)	Modifies sugars; its role is less clear

Next to them are regulatory DNA sequences — places where regulatory proteins can bind:

- The **promoter** — where RNA polymerase binds.
- The **operator** — a *negative* control site. When the lac repressor sits on it, RNA polymerase is blocked. It overlaps the promoter.
- The **CAP-binding site** — a *positive* control site. When CAP is bound, it helps RNA polymerase bind the promoter, so transcription goes up.
- The repressor itself is made from a separate gene, *lacI*, lying just upstream of the operon.

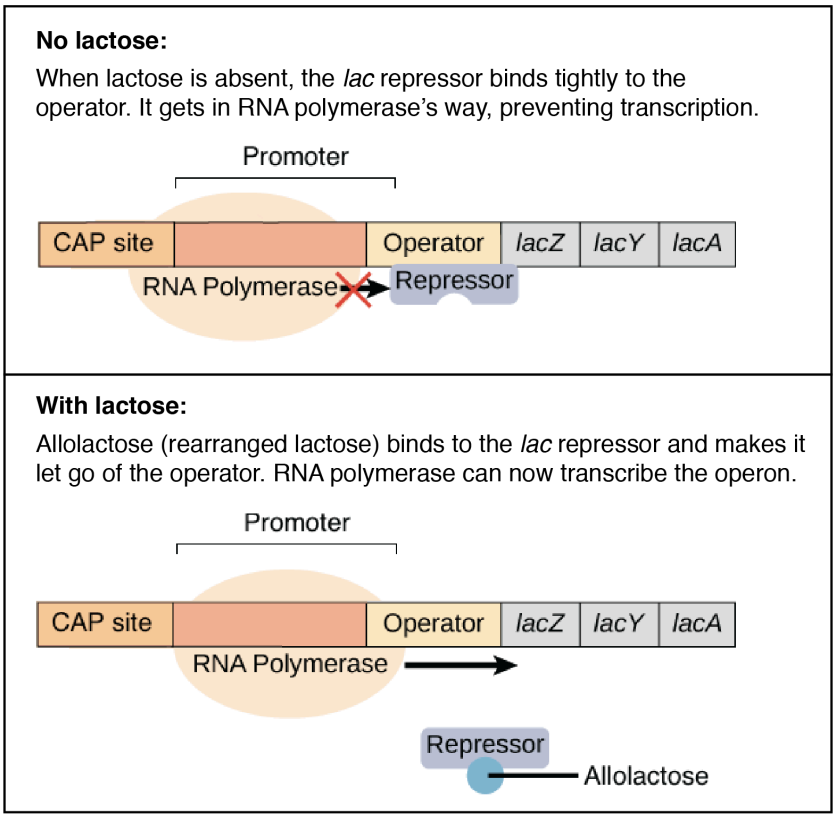


Figure 1. Top: with no lactose, the repressor sits on the operator and blocks RNA polymerase. Bottom: lactose is changed into allolactose, which binds the repressor and makes it let go, so RNA polymerase can transcribe the operon. *Source: Khan Academy*

The two controls

CONTROL 1 — THE LACTOSE SWITCH (NEGATIVE CONTROL)

- **No lactose:** the repressor binds the operator → transcription blocked.
- **Lactose present:** some lactose is converted to **allolactose**. Allolactose binds the repressor and changes its shape → it falls off the operator → RNA polymerase can pass.

CONTROL 2 — THE GLUCOSE SWITCH (POSITIVE CONTROL)

- **Glucose low:** the cell makes more of a small molecule called **cyclic AMP (cAMP)**. cAMP binds CAP, and the cAMP–CAP complex binds the CAP site and helps RNA polymerase start. Transcription goes **high**.
- **Glucose high:** cAMP is low, CAP does not bind, and transcription stays **low** even if lactose is there.

LINK TO DAY 3
In the lab we use a look-alike of allolactose called IPTG. It switches the operon on but the cell cannot break it down. We also use X-gal, a colourless substance that β-galactosidase turns blue. Together they give the blue–white screening you will use when cloning.

Worksheet: what is the expression level of the lac operon?

Fill in the table. For each situation, say whether the repressor is active or inactive, whether CAP is active or inactive, and what you expect the lac operon expression to be (none / very low / low / high).

Situation	Lac repressor	CAP	Lac operon expression
Lactose present Glucose absent			
Lactose present Glucose present			
Lactose absent Glucose absent			
Lactose absent Glucose present			

Think about it

1. A mutation in *lacI* means the repressor can no longer bind the operator. What happens to the operon in a medium with no lactose?
2. A mutation in the operator means the repressor can no longer recognise it. Same question.
3. A mutation means the repressor cannot bind allolactose. What will the operon do when lactose is present?
4. Why is it useful for *E. coli* to ignore lactose when glucose is plentiful?

ROLE-PLAY: "BE THE OPERON" (10 MINUTES)
Choose students to be RNA polymerase, the repressor, CAP, and a lactose molecule. Mark the DNA on the floor with tape: CAP site – promoter – operator – lacZ, lacY, lacA. Call out a scenario ("lactose present, glucose present") and let the group act it out: Does the repressor stand on the operator? Is CAP at its site? Does RNA polymerase walk through?

Repeat for all four scenarios, then compare with the worksheet.

3. Quorum sensing

Quorum sensing is the way bacteria communicate. Cells release small signal molecules into their surroundings. When there are only a few bacteria, the signal drifts away and stays at a very low level. When the population becomes dense — a quorum — the signal builds up. When it reaches a critical level it binds to receptor proteins in the cells and switches on a set of genes in all the cells at the same time.

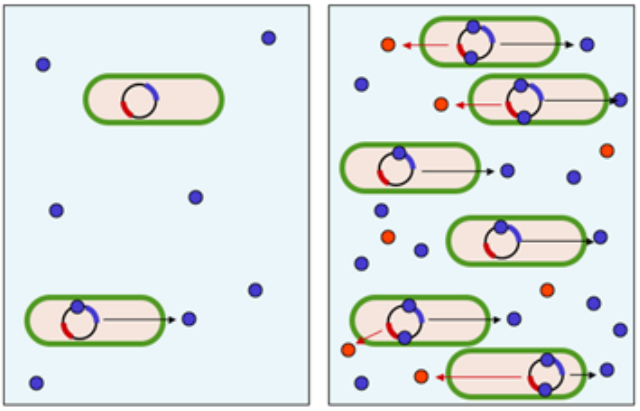


Figure 2. Left: few bacteria — the signal molecules (dots) are too dilute. Right: many bacteria — the signal reaches the threshold and the genes switch on. *Source: Wikimedia Commons*

How it works, step by step

1. Every cell makes a small amount of a signal molecule (an **autoinducer**) all the time.
2. The signal leaves the cell and spreads into the environment.
3. At low cell density the signal is diluted and stays below the threshold.
4. At high cell density the signal accumulates. It enters (or is detected by) the cells and binds a **receptor**.
5. The receptor switches on target genes — often including the gene that makes more signal. This **positive feedback** makes the whole population switch together.

	Gram-negative bacteria	Gram-positive bacteria
Signal	Small molecules, often AHLs (acyl-homoserine lactones)	Short peptides (autoinducing peptides)
Detected by	Receptor protein inside the cell (e.g. LuxR)	Receptor on the cell surface

Many species also release a "universal" signal called AI-2, which lets different species sense each other.

Examples

- *Vibrio fischeri* lives in the light organ of the Hawaiian bobtail squid. When the bacteria are packed densely in the organ, they make light (**bioluminescence**). In open sea water they are too few and stay dark.
- *Pseudomonas aeruginosa* uses quorum sensing to form **biofilms** and release toxins only when there are enough cells for a successful attack. Biofilms are sticky communities that protect bacteria from drugs.

- **Waiting until you are many** helps pathogens avoid alerting the immune system too early.

Why it matters

- **Medicine:** drugs that block quorum sensing ("**quorum quenching**") could disarm bacteria without killing them, which puts less pressure on them to become resistant.
- **Synthetic biology:** scientists build quorum-sensing circuits so that engineered bacteria act together, for example to produce a product only once a culture is dense enough.

GAME: "THE BEAD QUORUM" (10 MINUTES)

Each student is a bacterium holding a few beads (signal molecules). A bowl in the middle is the environment. In each round every student adds 1 bead to the bowl.

Round A — spread out: stand far apart (use a large room). Only 5 students play; the bowl is a "large lake", so remove 2 beads after each round to show dilution. Can you reach the threshold of 20 beads?

Round B — packed together: all students stand in a tight circle with no dilution. How many rounds until 20? When the threshold is reached, everybody stands up and shouts "GLOW!".

Discuss: what does the threshold tell the bacteria about their population?

FURTHER RESOURCE

HHMI BioInteractive has a free animation and classroom resource on bacterial quorum sensing:
https://www.biointeractive.org/classroom-resources/bacterial-quorum-sensing

Teacher answer key (worksheet)

Note: the earlier version of this worksheet said "medium" for the second row. Most textbooks say low, because without CAP help the promoter is weak even though the repressor is not blocking it.

Answers: think about it

1. The operon is on all the time (**constitutive**), even without lactose — a waste of energy.
2. Same: the repressor cannot block, so the operon is on all the time.
3. The repressor stays on the operator, so the operon stays **off** even with lactose (the cell cannot use lactose).
4. Glucose gives more energy more quickly. Making lac enzymes when glucose is plentiful would waste energy and materials.